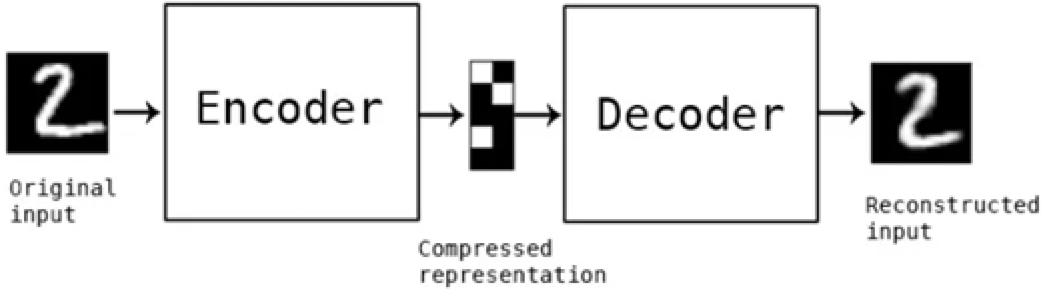
Autoencoders

Introduction

An autoencoder is a type of neural network architecture designed to efficiently compress (encode) input data down to its essential features, then reconstruct (decode) the original input from this compressed representation.

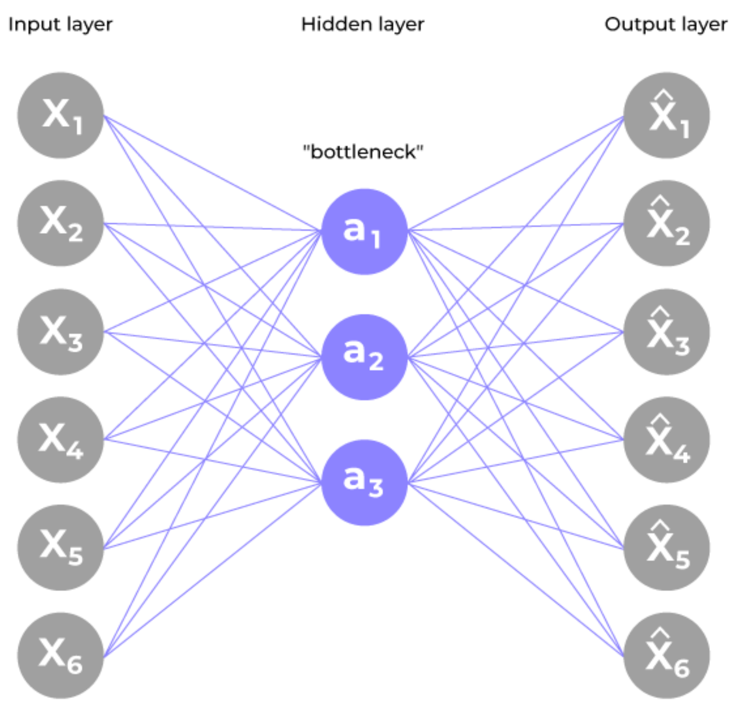


Autoencoding is training a network to replicate its input to its output

Main Architecture of Autoencoder

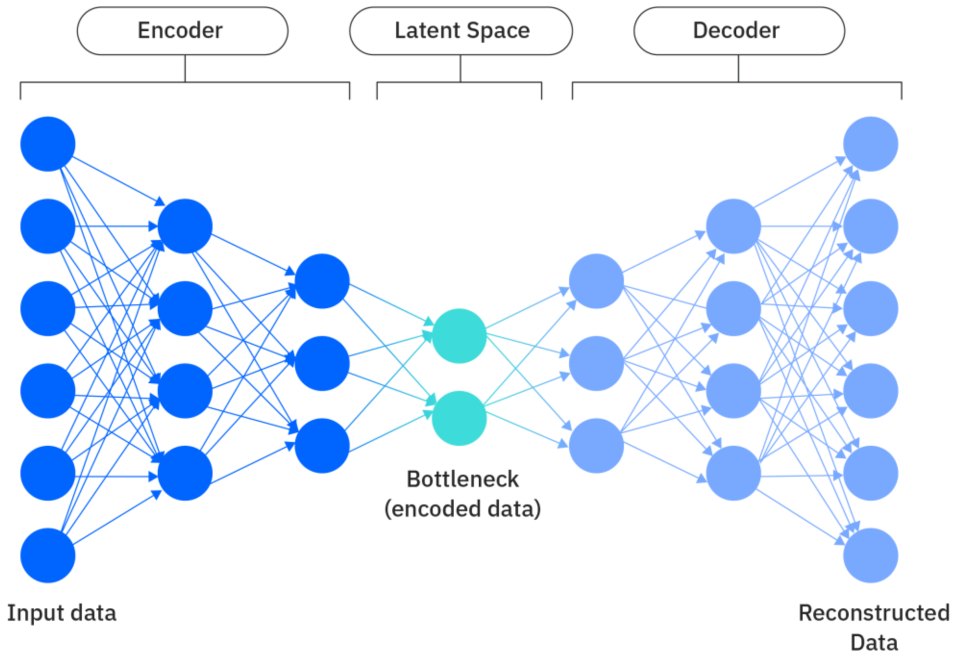
An autoencoder typically consists of three blocks

- **Encoder layer to compress the input data into a compressed representation.**
- **Bottleneck layer or code to represent the compressed input.**
- **Decoder layer to reconstruct the encoded image back to the original dimension.**



Encoder

- Input layer take raw input data
- The hidden layers progressively reduce the dimensionality of the input, capturing important features and patterns. These layers compose the encoder.



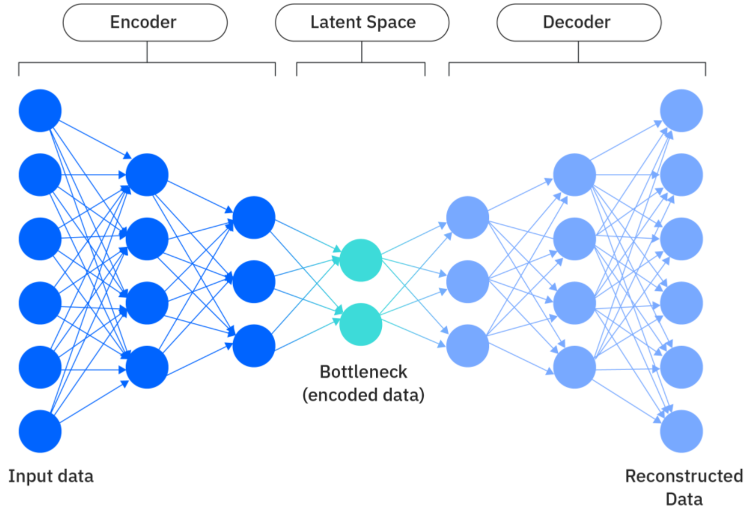
bottleneck layer

From the encoder point of view:

- The bottleneck layer (latent space) is the final hidden layer, where the dimensionality is significantly reduced. This layer represents the compressed encoding of the input data.

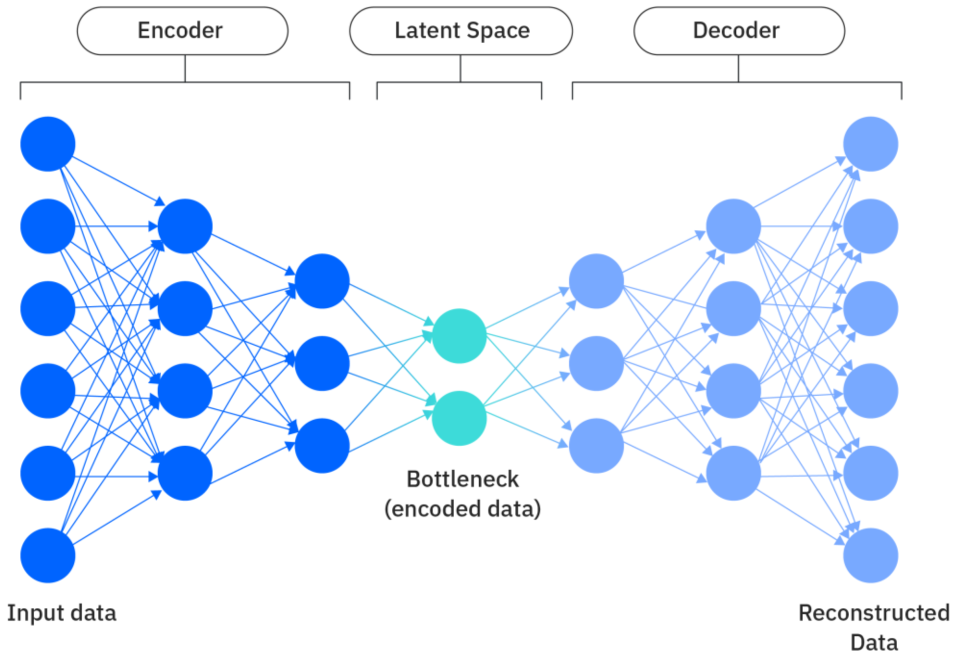
From the decoder point of view:

- The bottleneck layer takes the encoded representation and expands it back to the dimensionality of the original input.



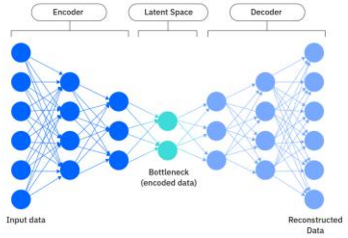
Decoder

- The hidden layers progressively increase the dimensionality and aim to reconstruct the original input.
- The output layer produces the reconstructed output, which ideally should be as close as possible to the input data.



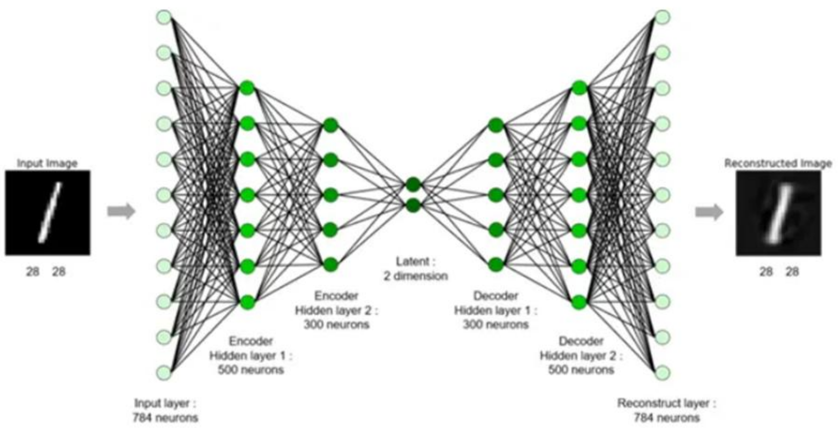
The loss function

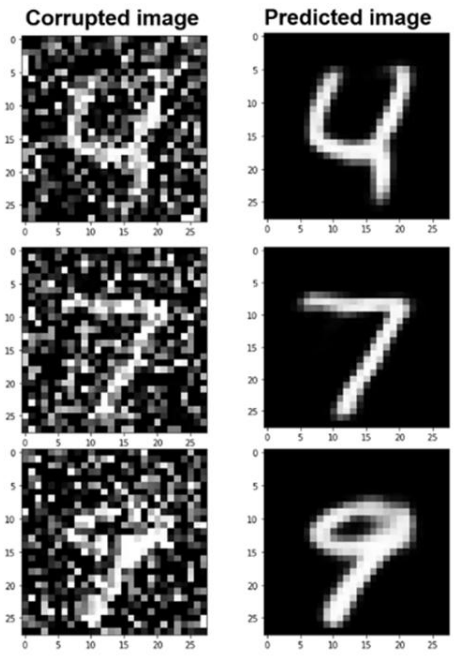
- The loss function used during training is typically a reconstruction loss, measuring the difference between the input and the reconstructed output. Common choices include
  - mean squared error (MSE) for continuous data
  - binary cross-entropy for binary data.
- During training, the autoencoder learns to minimize the reconstruction loss, forcing the network to capture the most important features of the input data in the bottleneck layer.



Types of Autoencoders
1- Denoising Autoencoder

- Denoising autoencoder works on a partially corrupted input and trains to recover the original undistorted image. As mentioned above, this method is an effective way to constrain the network from simply copying the input and thus learn the underlying structure and important features of the data.





Types of Autoencoders
1- Denoising Autoencoder

Advantages

1. This type of autoencoder can extract important features and reduce the noise or the useless features.

2. Denoising autoencoders can be used as a form of data augmentation, the restored images can be used as augmented data thus generating additional training samples.

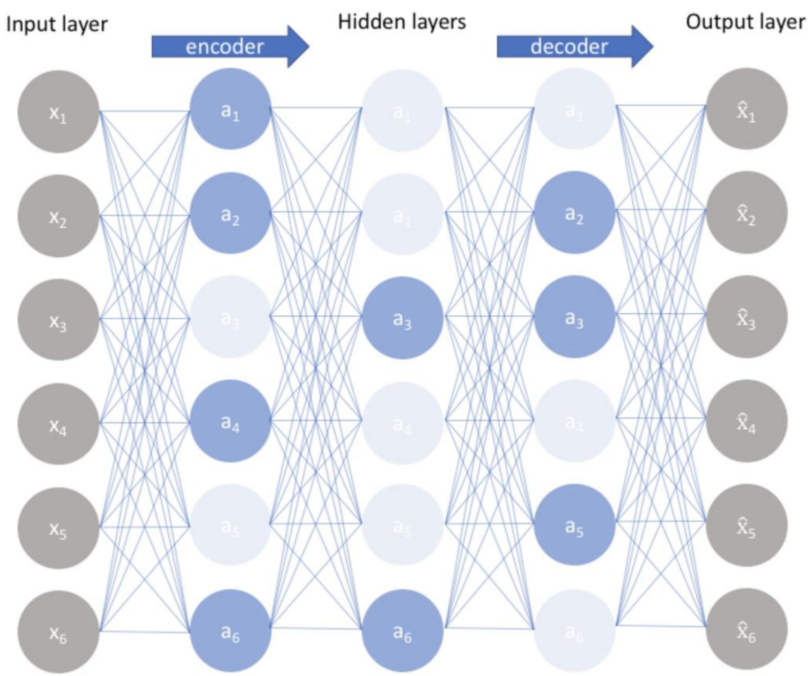
Disadvantages

1. Selecting the right type and level of noise to introduce can be challenging and may require domain knowledge.

2. Denoising process can result into loss of some information that is needed from the original input. This loss can impact accuracy of the output.

Types of Autoencoders
2- Sparse Autoencoder

- This type of autoencoder typically contains more hidden units than the input but only a few are allowed to be active at once. This property is called the sparsity of the network. The sparsity of the network can be controlled by either manually zeroing the required hidden units, tuning the activation functions or by adding a loss term to the cost function.



Types of Autoencoders
2- Sparse Autoencoder

Advantages

1. The sparsity constraint in sparse autoencoders helps in filtering out noise and irrelevant features during the encoding process.

2. These autoencoders often learn important and meaningful features due to their emphasis on sparse activations.

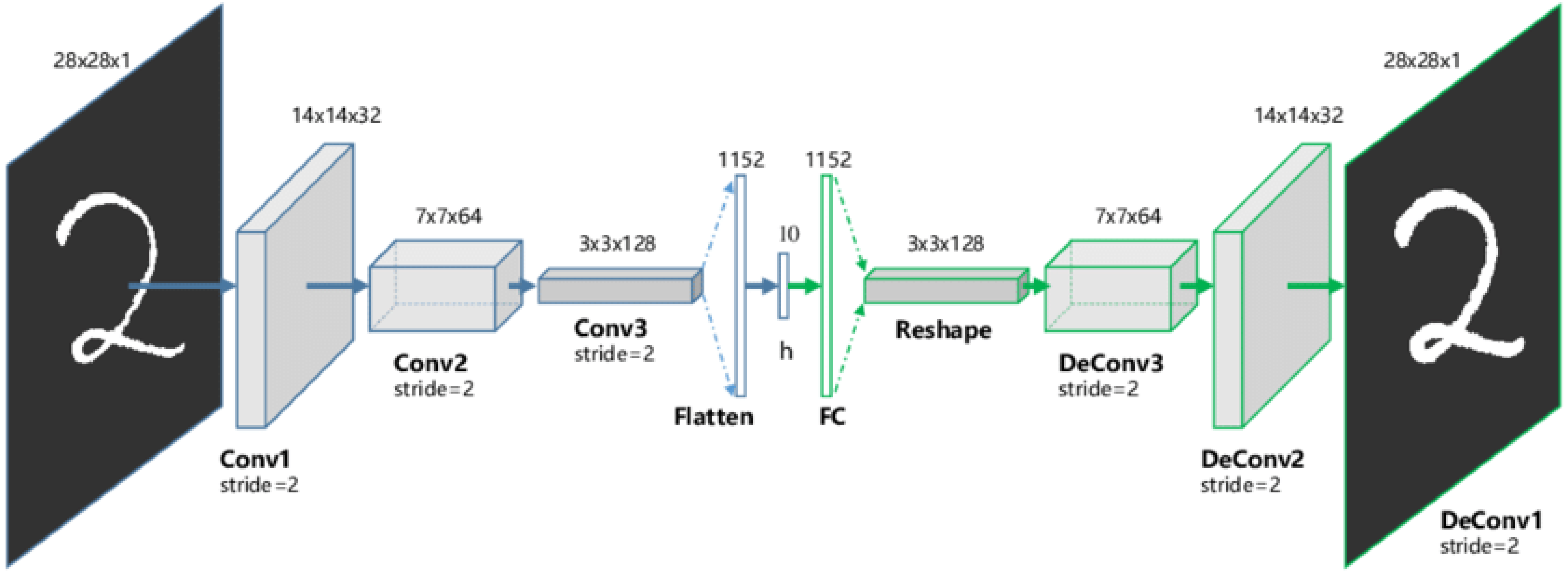
Disadvantages

1. The choice of hyperparameters play a significant role in the performance of this autoencoder. Different inputs should result in the activation of different nodes of the network.

2. The application of sparsity constraint increases computational complexity.

Types of Autoencoders
3- Convolutional Autoencoder

- Convolutional autoencoders are a type of autoencoder that use convolutional neural networks (CNNs) as their building blocks. The encoder consists of multiple layers that take a image or a grid as input and pass it through different convolution layers thus forming a compressed representation of the input. The decoder is the mirror image of the encoder it deconvolves the compressed representation and tries to reconstruct the original image.



Types of Autoencoders
3- Convolutional Autoencoder

Advantages

1. Convolutional autoencoder can compress high-dimensional image data into a lower-dimensional data. This improves storage efficiency and transmission of image data.

2. Convolutional autoencoder can reconstruct missing parts of an image. It can also handle images with slight variations in object position or orientation.

Disadvantages

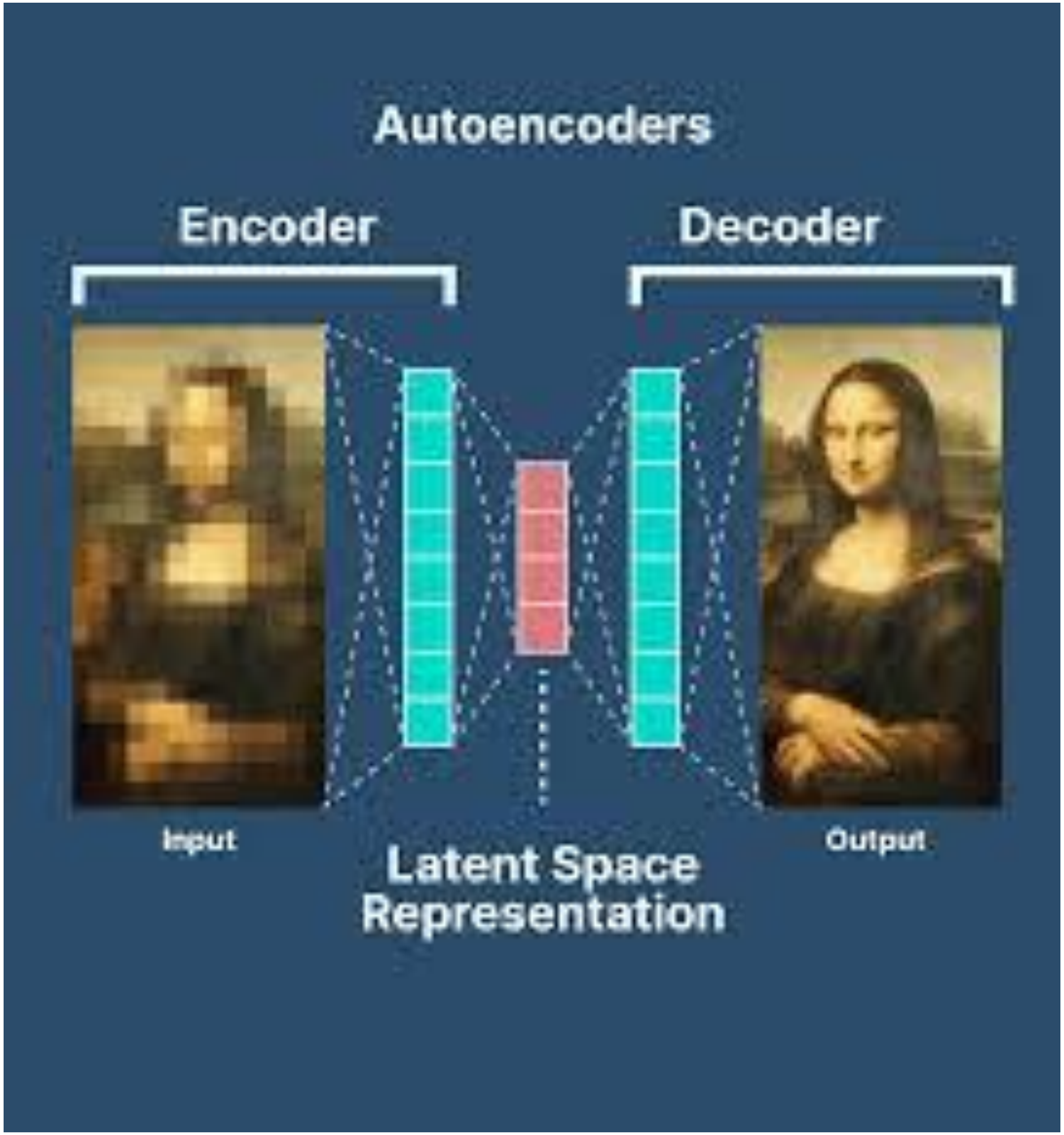
1. These autoencoder are prone to overfitting. Proper regularization techniques should be used to tackle this issue.

2. Compression of data can cause data loss which can result in reconstruction of a lower quality image.

Applications of AE

Auto-Encoders are designed to learn a lower-dimensional representation for a higher-dimensional data. Applications include

1. Dimensionality Reduction
2. Feature Extraction
3. Image Denoising
4. Image Compression
5. Image Search
6. Anomaly Detection
7. Missing Value Imputation


Autoencoders
Encoder
Decoder
Input
Output
Latent Space
Representation

Autoencoders vs Encoder-decoders

Though all autoencoders include both an encoder and a decoder, not all encoder-decoder models are autoencoders.

Encoder-decoder frameworks are used in a variety of deep learning models to extracts/encode features of the input data and take the extracted feature data to the decoder for tasks such as classification or segmentation. **In applications of such encoder-decoder models, the output of the neural network is different from its input.**

For example, in segmentation models like U-Net, the encoder extracts features from the input to determine pixel classification; using the feature map and those pixel-wise classifications, the decoder then constructs segmentation masks for each object in the image. The goal of these encoder-decoder models is to label pixels by their semantic class. they are trained via supervised learning, optimizing the model’s predictions against ground truth images.

Autoencoders

- Autoencoders form a very specific subset of encoder-decoder architectures that are trained via unsupervised learning to reconstruct their own input data.
- Input and output dimensions are the same.
- Using unsupervised machine learning, autoencoders are trained to discover the latent variables of the input data.
- The latent variables are not directly observable but they fundamentally inform the way data is distributed. Collectively, the latent variables of a given set of input data are referred to as a latent space.